

Supplementary Materials

Supplementary Methods

Geometric morphometric

We identified five landmarks (homologous points) for the posterior sepal: 1) the tip of the upper petal, 2) the insertion of the pedicel on the perianth, 3) the extremity of the spur, 4) the lower extremity of the opening of the posterior sepal, and 5) the lower point of the lower petal closer to the camera (Figure 5A). We also defined two curves with 30 semilandmarks each to capture the shape of the posterior sepal. The first curve followed the outline of the posterior sepal between landmarks 2 and 3 (dorsal curve, Figure 5A), and the second one followed the outline of the posterior sepal between landmarks 3 and 4 (ventral curve, Figure 5A).

We placed seven landmarks for the front views of the flowers: 1) the tip of the upper petal, 2) the intersection between the upper petal and the right lateral petal (left for landmark 6, which is symmetrical), 3) the right widest point of the corolla opening (left for landmark 5, which is symmetrical), 4) the intersection of the two lower petals or, if the two petals do not intersect, the center of a segment drawn between the uppermost part of each lower petal, and, 7) the lower point of the stamens or the stigma (Figure 5D). Note that points 4 and 7 are better defined as semi-homologous landmarks. We also defined four semilandmarks curves to capture the shape of the corolla. The first and fourth followed the outline of the upper petal between landmarks 1 and 2, or 6 and 1, and each contained 10 points. The second and third contained 35 points and followed the lateral and lower petal outline between landmarks 2 and 4, or 4 and 6 (Figure 5D). Curves were numbered clockwise starting from the top of the flower.

We removed flower pictures that were too blurry or for which critical features of the flowers were masked, preventing us from precisely locating landmarks or semilandmarks. For all curves, we removed the first and last semilandmarks because they were redundant with the landmarks. For profile pictures, we also removed landmarks 1 and 5 because they were redundant with the landmarks and semilandmarks of the front pictures and were sometimes challenging to locate precisely. Different persons placed the landmarks, but one person (JB) validated all points and curves of all flowers.

For the profile view, the sepal size corresponded to the area of a polygon created by concatenating all curves and landmarks 2, 3, and 4 (Figure 3A, Figure 5A). For the front view, the

corolla surface corresponded to the area of a polygon created by concatenating all curves and landmarks 1, 2, 4 and 6 (Figure 3E, Figure 5D).

Supplementary Figures

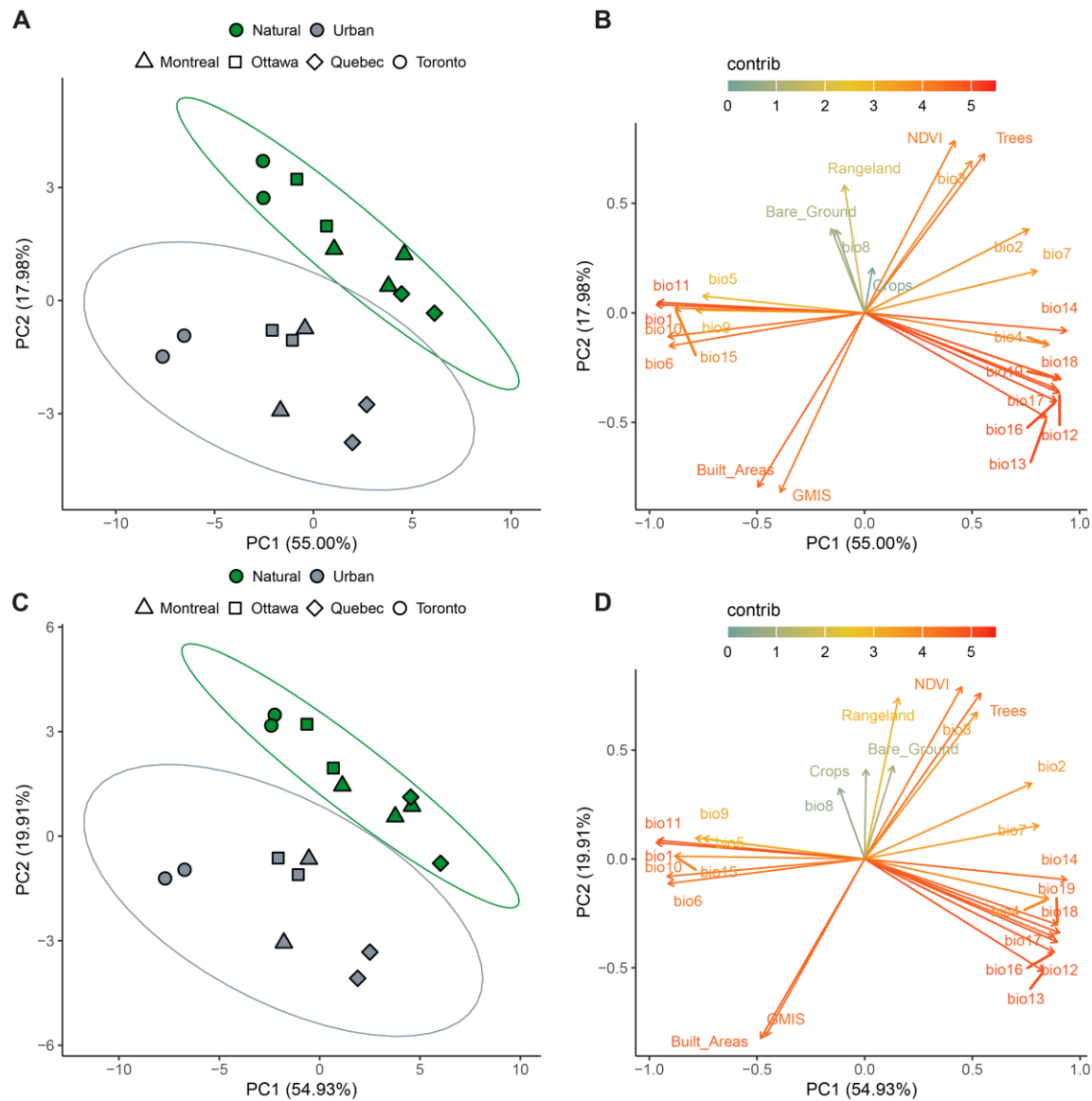


Figure S1. Population characteristics at different spatial scales. (A-B) Individual and variables plot of a Principal Component Analysis (PCA) on urbanization and bioclimatic variables computed using a circle of 1000m radius around each population. (C-D) Individual and variables plot of a Principal Component Analysis (PCA) on urbanization and bioclimatic variables computed using a circle of 2000m radius around each population. Urban and natural populations mainly differ in the percentage of impervious surfaces, trees and isothermality (BIO3) in both analyses, whereas cities vary according to temperatures and precipitations.

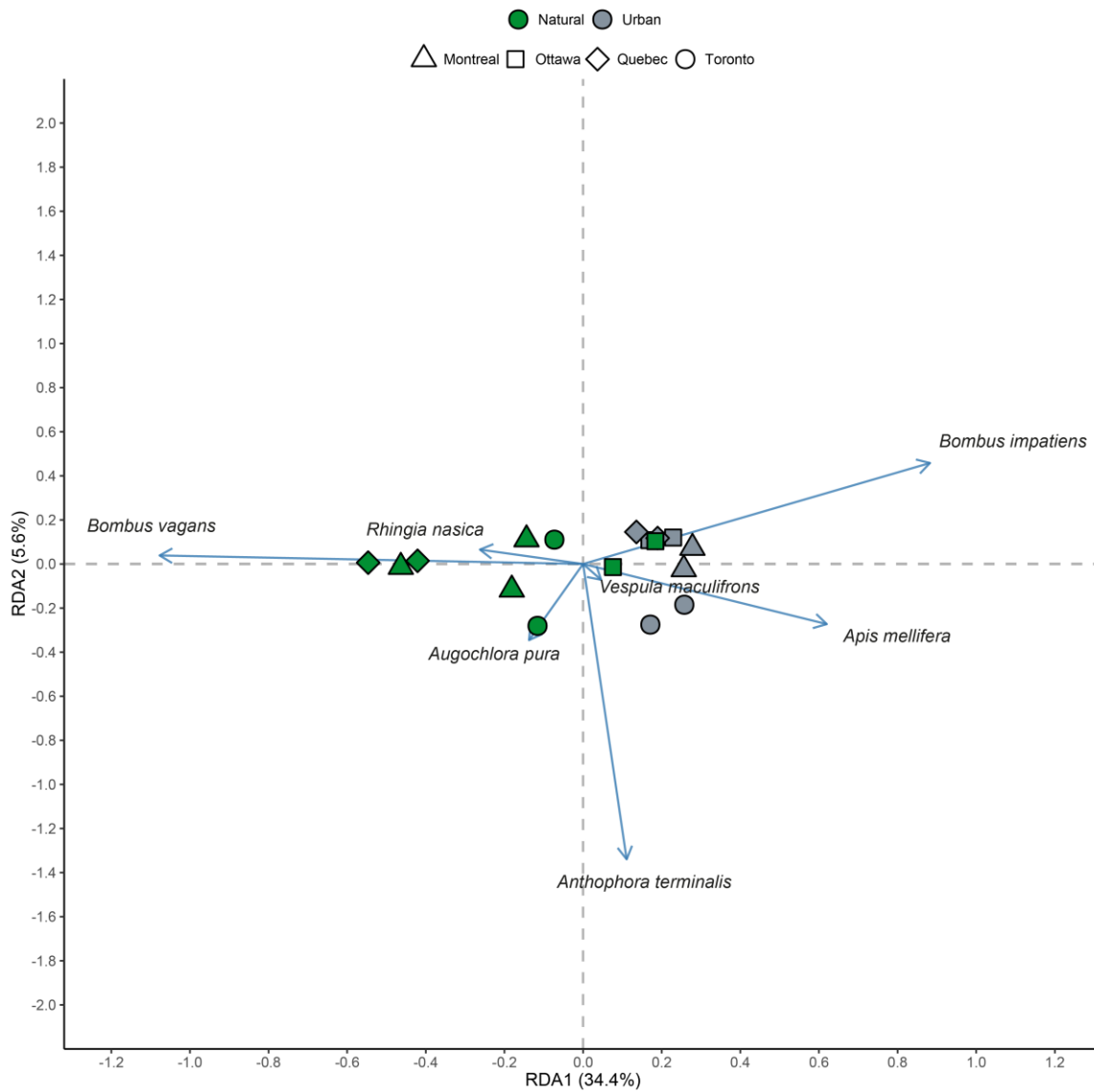


Figure S2. Redundancy analysis (RDA) evaluating the impact of urbanization on the *Impatiens capensis* pollinator community in 2021 and 2022. When both years are considered together, the main difference between urban and natural habitats is the bumblebee species pollinating *I. capensis*. *Anthophora terminalis* separates the Toronto region because it was overrepresented in this region in 2021.

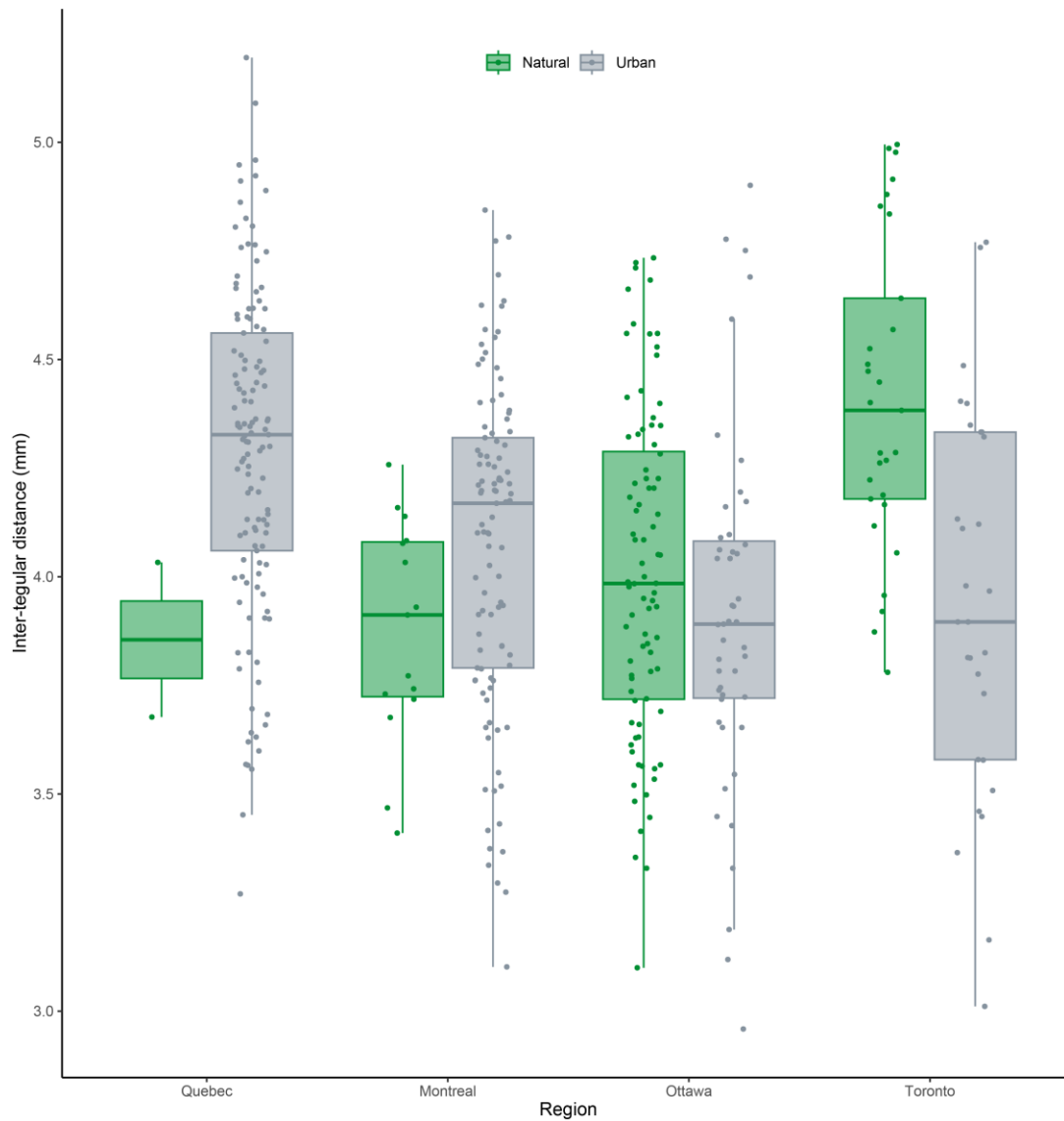


Figure S3. Boxplots of *Bombus impatiens* inter-tegular distances by habitat and region, averaged over the two field seasons. Dots represent individual bumblebee measurements.

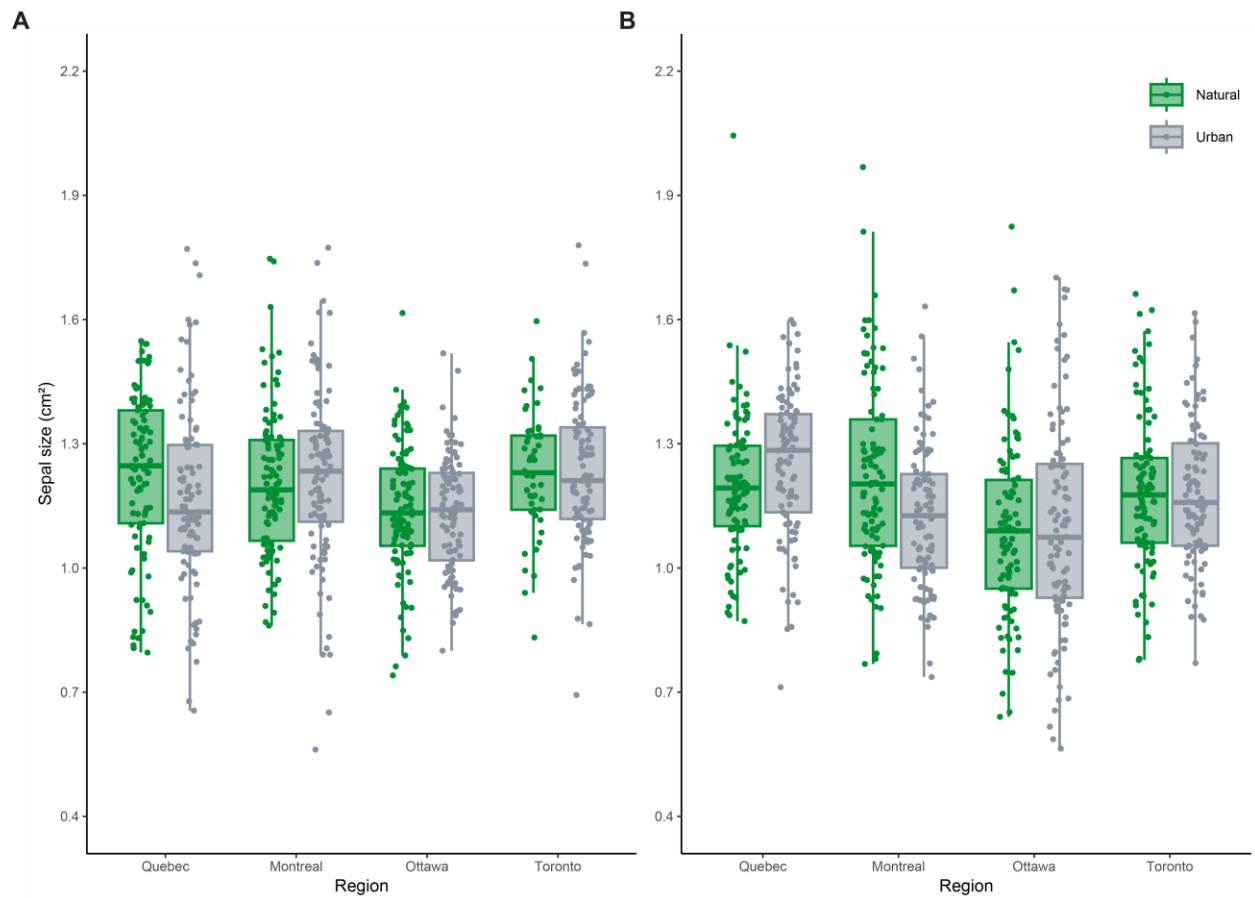


Figure S4. Sepal size in the field according to region and habitat type for 2021 (A) and 2022 (B). See Table S1 for more information about sampling and Table S6 for analysis of sepal size variation.

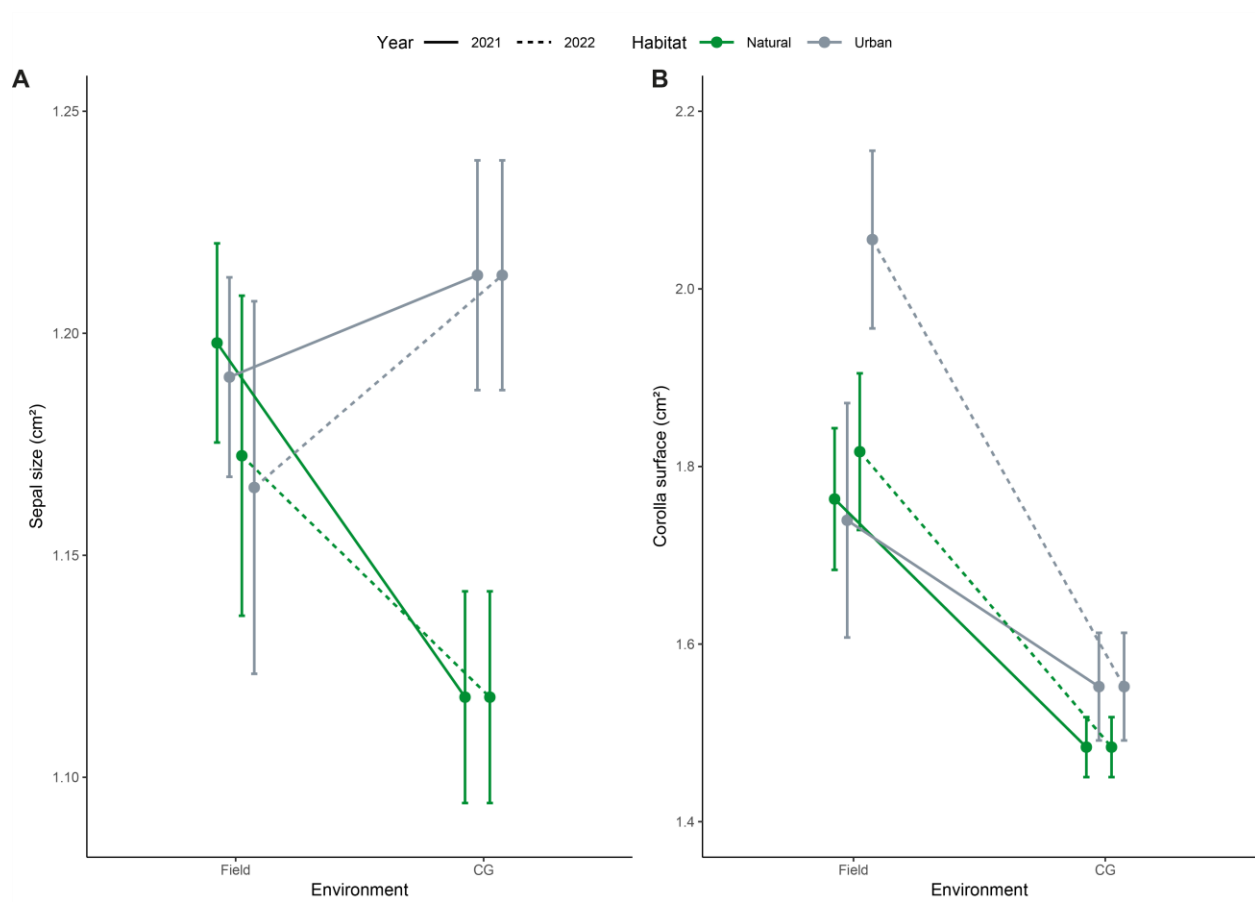


Figure S5. Mean response of sepal size (A) and corolla surface (B) to environment change, grouped by habitat. Solid lines represent 2021 data in the field compared to the unique common garden experiment, and dashed lines represent 2022 data in the field compared to the unique common garden experiment.

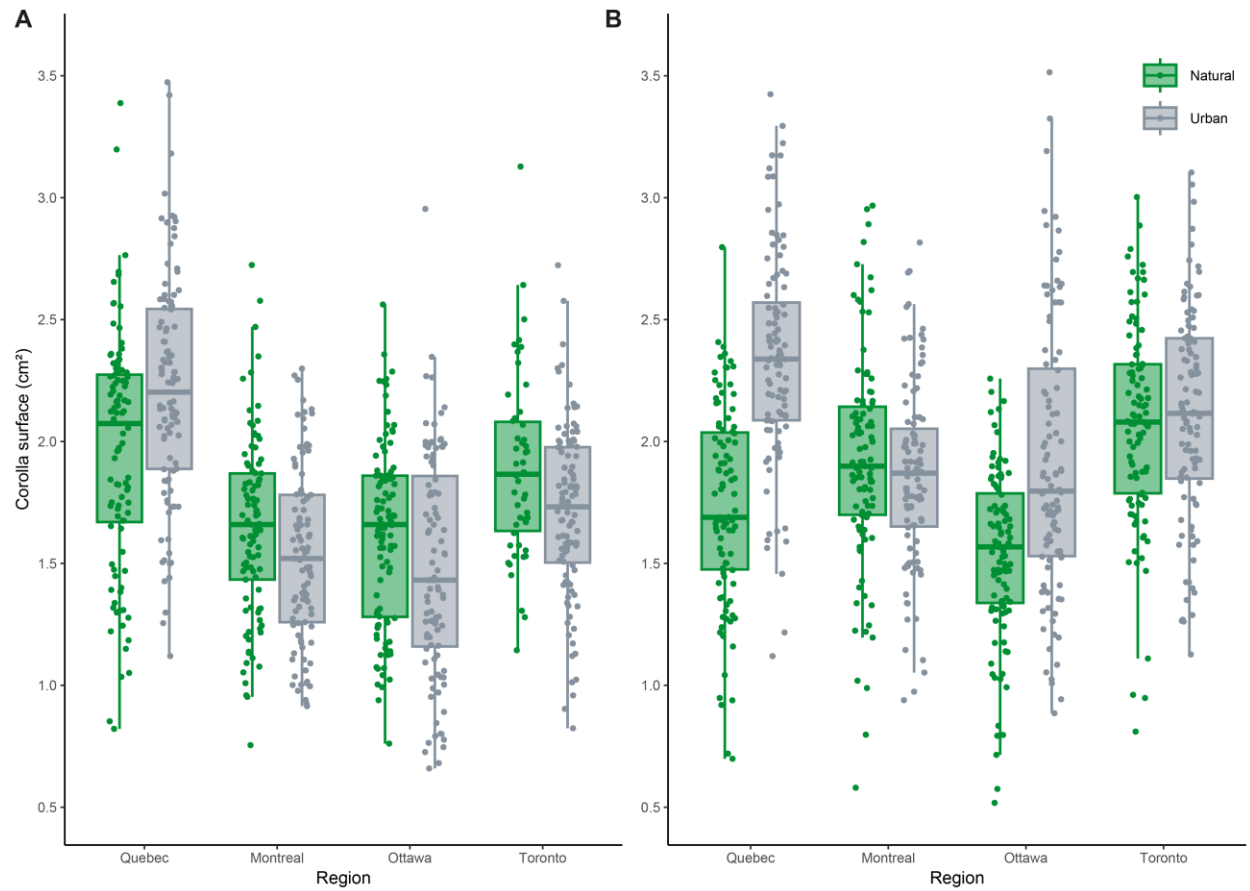


Figure S6. Corolla surface in the field according to region and habitat type for 2021 (A) and 2022 (B). See Table S1 for more information about sampling and Table S6 for analysis of corolla surface variation.

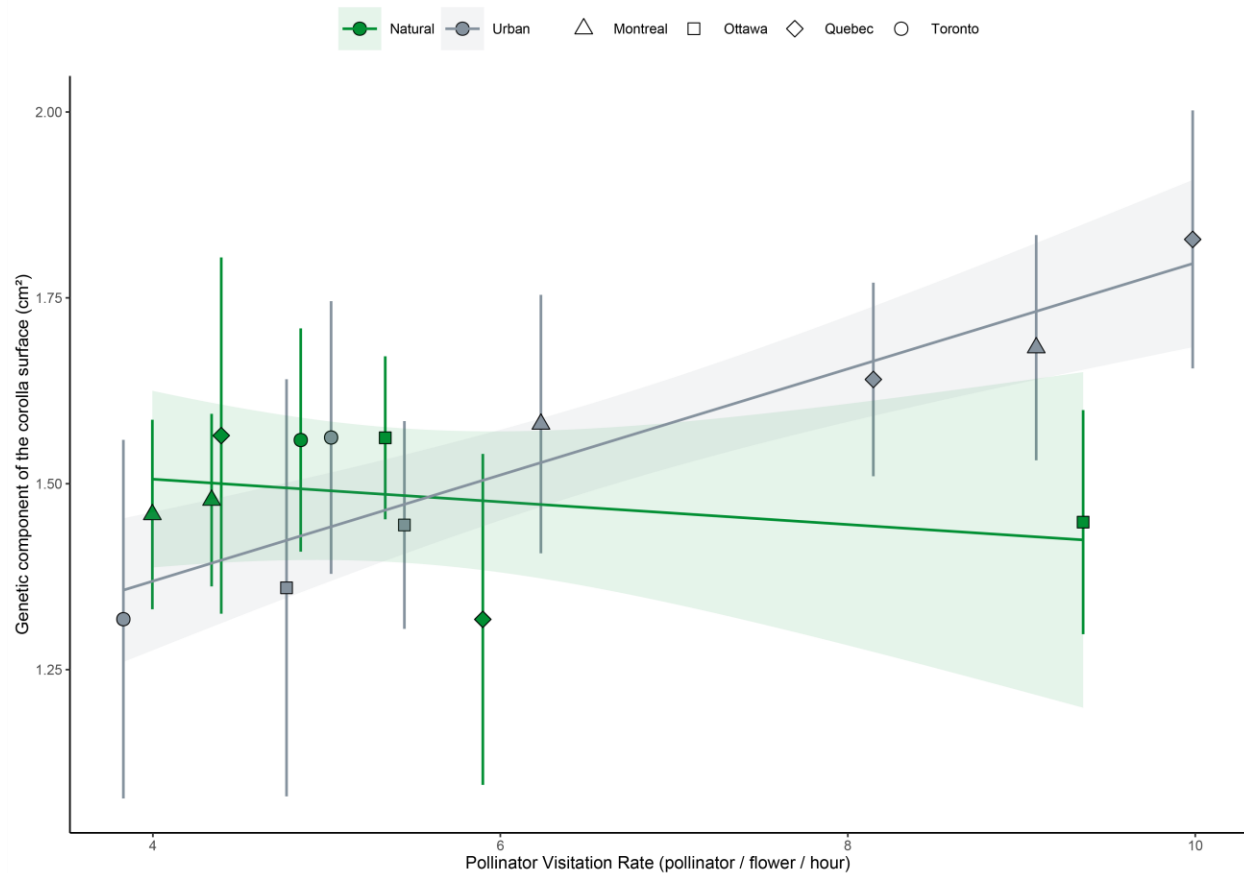


Figure S7. Correlation between the genetic component of the corolla surface and the pollinator visitation rate. There was a significant effect of the pollinator visitation rate on corolla surface, and a significant interaction between pollinator visitation rate and habitat. Regression lines for urban and natural populations are shown, with their standard errors. The points represent the population mean of the corolla surface in the common garden experiment (with 95% confidence intervals of the mean) and the population mean of the pollinator visitation rate averaged across the 2021 and 2022 field seasons.

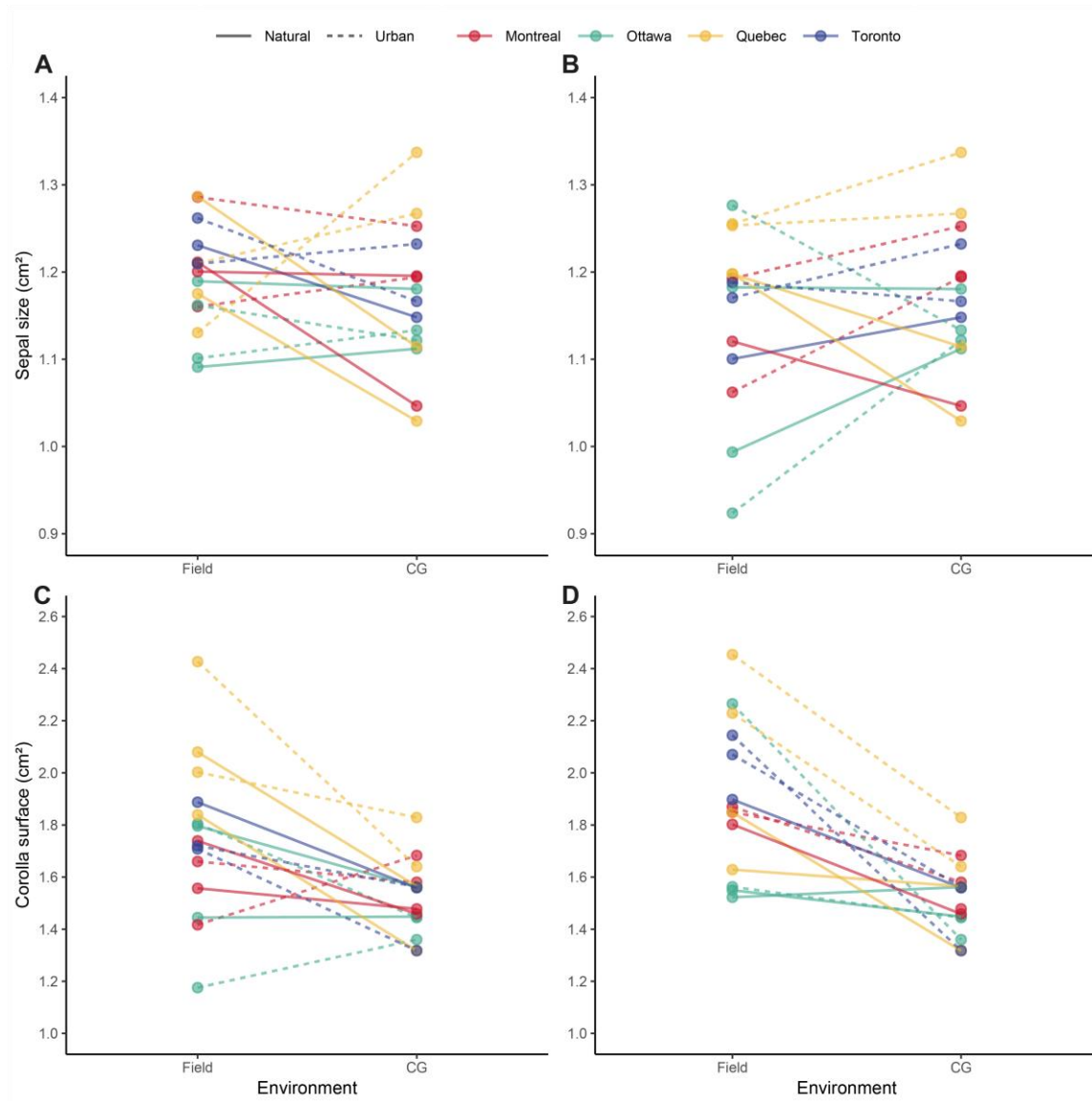


Figure S8. Plastic response of sepal size and corolla surface, grouped by region. Lines represent populations colored according to their region. Solid lines represent natural populations, and dashed lines represent urban populations. (A) Sepal size response for 2021 field data vs common garden data. (B) Sepal size response for 2022 field data vs common garden data. (C) Corolla surface response for 2021 field data vs common garden data. (D) Corolla surface response for 2022 field data vs common garden data.

Supplementary Tables

Table S1. Populations studied. The first number in the sample size column is for the 2021 field season, the second one for the 2022 field season, and the third for the common garden experiment. The first line in the sample size column for each population is for profile pictures (sepal), and the second one is for front pictures (corolla).

Population	Region	Habitat	Latitude	Longitude	Sample size
Bergeron (BER)	Montreal	Urban	45.651	-73.753	50/50/26 44/50/27
Bois Franc (BFC)	Montreal	Urban	45.505	-73.708	49/49/28 49/49/28
Mirabel (MIR)	Montreal	Natural	45.683	-74.213	NA/50/NA NA/50/NA
Oka (OKA)	Montreal	Natural	45.482	-74.035	49/NA/39 48/NA/39
SBL (SBL)	Montreal	Natural	45.974	-74.020	50/50/21 49/49/21
Beaver Dam (DAM)	Ottawa	Natural	45.293	-75.823	50/49/25 50/50/25
Gatineau (GAT)	Ottawa	Natural	45.474	-75.798	50/50/14 50/50/14
Hampton Park (HAM)	Ottawa	Urban	45.391	-75.740	50/50/27 47/50/27
Riverain Park (RIV)	Ottawa	Urban	45.429	-75.670	50/50/13 49/50/13
Maizerets (MAI)	Quebec	Urban	46.838	-71.211	50/50/30 50/50/28
Marais du Nord (MDN)	Quebec	Natural	46.963	-71.388	47/50/17 48/50/17
Saules (SAU)	Quebec	Urban	46.81	-71.316	47/50/25 50/50/25
Savane (SAV)	Quebec	Natural	46.633	-71.287	50/49/12 48/49/12
KSR (KSR)	Toronto	Natural	44.029	-79.528	50/50/24 50/50/24
Silver Creek (SCK)	Toronto	Natural	43.703	-79.964	NA/50/NA NA/50/NA
TCS (TCS)	Toronto	Urban	43.701	-79.328	49/50/9 50/50/9
YCP (YCP)	Toronto	Urban	43.682	-79.383	50/50/22 50/50/22

Table S2. Pollinator relative frequency calculated on collected specimens. The 2022 collection effort was more important, and this explains why more specimens were observed. NA are for species not found in either 2021 or 2022. Values are rounded.

Order	Family	Species	Frequency		
			2021 (n = 171 insects)	2022 (n = 521 insects)	Cumulated (n=692 insects)
Diptera	Calliphoridae	<i>Calliphoridae</i>	NA	0.004	0.003
Diptera	Muscidae	<i>Muscidae</i>	NA	0.002	0.001
Diptera	Syrphidae	<i>Epistrophe sp.</i>	NA	0.002	0.001
Diptera	Syrphidae	<i>Platycheirus nectaricus</i>	NA	0.013	0.010
Diptera	Syrphidae	<i>Rhingia nasica</i>	NA	0.019	0.014
Diptera	Syrphidae	<i>Toxomerus geminatus</i>	NA	0.004	0.003
Diptera	Syrphidae	<i>Unknown Syrphidae</i>	NA	0.002	0.001
Hymenoptera	Apidae	<i>Anthophora terminalis</i>	0.070	0.036	0.045
Hymenoptera	Apidae	<i>Apis mellifera</i>	0.088	0.056	0.064
Hymenoptera	Apidae	<i>Bombus borealis</i>	0.012	NA	0.003
Hymenoptera	Apidae	<i>Bombus fervidus</i>	0.006	0.002	0.003
Hymenoptera	Apidae	<i>Bombus impatiens</i>	0.596	0.633	0.624
Hymenoptera	Apidae	<i>Bombus ternarius</i>	0.018	0.002	0.006
Hymenoptera	Apidae	<i>Bombus vagans</i>	0.099	0.136	0.127
Hymenoptera	Apidae	<i>Ceratina calcarata</i>	NA	0.002	0.001
Hymenoptera	Crabronidae	<i>Ectemnius continuus</i>	NA	0.004	0.003
Hymenoptera	Halictidae	<i>Agapostemon virescens</i>	NA	0.002	0.001
Hymenoptera	Halictidae	<i>Augochlora pura</i>	0.018	0.015	0.016
Hymenoptera	Megachilidae	<i>Megachile mendica</i>	0.006	0.002	0.003
Hymenoptera	Megachilidae	<i>Megachile relativa</i>	NA	0.002	0.001
Hymenoptera	Vespidae	<i>Ancistrocerus diabatus</i>	NA	0.002	0.001
Hymenoptera	Vespidae	<i>Dolichovespula renaria</i>	0.023	0.012	0.014
Hymenoptera	Vespidae	<i>Unknown Vespu</i>	0.006	NA	0.001
Hymenoptera	Vespidae	<i>Vespula acadica</i>	0.006	NA	0.001
Hymenoptera	Vespidae	<i>Vespula alascensis</i>	NA	0.002	0.001
Hymenoptera	Vespidae	<i>Vespula consobrina</i>	0.041	0.004	0.013
Hymenoptera	Vespidae	<i>Vespula flavipila</i>	NA	0.002	0.001
Hymenoptera	Vespidae	<i>Vespula germanica</i>	NA	0.004	0.003
Hymenoptera	Vespidae	<i>Vespula maculifrons</i>	0.012	0.036	0.030

Table S3. Analysis of the pollinator visitation rate variation

A) Analysis of variance table for Pollinator Visitation Rate				
Effect	Pollinator Visitation Rate			
	Sum of Squares	df	F	P
Region	19.629	3	1.500	0.241
Habitat	3.773	1	0.865	0.362
Region $\times$ Habitat	48.087	3	3.675	0.027
Residuals	100.316	23		

B) Coefficients of the model				
	Estimate	Std Error	t	P
Global mean	6.250	0.377	16.587	
Montreal	0.376	0.644	0.584	0.565
Ottawa	-0.023	0.644	-0.036	0.971
Quebec	0.856	0.644	1.330	0.197
Toronto	-1.209	0.678	-1.783	0.088
Natural	-0.315	0.377	-0.837	0.411
Montreal $\times$ Natural	-0.718	0.644	-1.115	0.276
Ottawa $\times$ Natural	1.434	0.644	2.227	0.036
Quebec $\times$ Natural	-1.645	0.644	-2.554	0.018
Toronto $\times$ Natural	0.928	0.678	1.369	0.184

C) Estimated marginal means contrasts (Urban - Natural)				
Urban - Natural contrast	Estimate	t	df	P
Quebec	3.92	2.654	23	0.057
Montreal	2.07	1.399	23	0.430
Ottawa	-2.24	-1.515	23	0.430
Toronto	-1.23	-0.769	23	0.450

All values are rounded to the third decimal. **A)** Analysis of variance table **B)** Coefficients of the model extracted from the summary. **C)** We estimated the effect of habitat on pollinator-visitation rates for each region using post-hoc tests comparing estimated marginal means contrasts. We corrected for multiple comparisons using a holm correction. Note that the P-values are computed with type II sum of squares in A and with type III sum of squares in B.

Table S4. Sample size and summary of bumblebee sizes. This table summarizes the 2021 and 2022 sampling years combined. n is the sample size for each population, and N is the sample size for each habitat within a region. All bumblebee species are combined.

Population	Habitat	Region	n	ITD	sd	se	ci	N
MDN	Natural	Quebec	19	3.857	0.402	0.092	0.194	57
SAV	Natural	Quebec	38	3.796	0.324	0.053	0.106	
MAI	Urban	Quebec	63	4.473	0.33	0.042	0.083	128
SAU	Urban	Quebec	65	4.101	0.328	0.041	0.081	
MIR	Natural	Montreal	19	3.828	0.239	0.055	0.115	34
OKA	Natural	Montreal	7	3.821	0.311	0.118	0.288	
SBL	Natural	Montreal	8	3.804	0.333	0.118	0.278	102
BER	Urban	Montreal	30	4.071	0.238	0.043	0.089	
BFC	Urban	Montreal	72	4.07	0.423	0.05	0.099	87
DAM	Natural	Ottawa	45	4.043	0.371	0.055	0.112	
GAT	Natural	Ottawa	42	3.953	0.368	0.057	0.115	49
HAM	Urban	Ottawa	16	4.178	0.345	0.086	0.184	
RIV	Urban	Ottawa	33	3.782	0.369	0.064	0.131	42
KSR	Natural	Toronto	6	3.994	0.356	0.145	0.374	
SCK	Natural	Toronto	36	4.331	0.382	0.064	0.129	29
TCS	Urban	Toronto	8	4.164	0.387	0.137	0.323	
YCP	Urban	Toronto	21	3.858	0.45	0.098	0.205	

Table S5. Analysis of the bumblebee size variation.

A) Wald χ^2 test				
Effect	Inter-Tegular Distance (ITD, mm)			P
	χ^2	df		
Year	0.109	1		0.740
Region	2.075	3		0.557
Habitat	2.415	1		0.120
Year $\times$ Region	1.990	3		0.574
Year $\times$ Habitat	0.302	1		0.582
Region $\times$ Habitat	8.827	3		0.032
Year $\times$ Region $\times$ Habitat	1.494	3		0.683

B) Coefficients of the model				
	Estimate	Std Error	t	P
Global mean	4.011	0.048	83.292	
2021	-0.017	0.031	-0.544	0.587
Montreal	-0.068	0.076	-0.894	0.393
Ottawa	-0.040	0.078	-0.514	0.620
Quebec	0.067	0.078	0.865	0.410
Toronto	0.041	0.099	0.414	0.683
Natural	-0.047	0.048	-0.984	0.345
2021 $\times$ Montreal	-0.020	0.050	-0.399	0.691
2021 $\times$ Ottawa	-0.004	0.040	-0.095	0.924
2021 $\times$ Quebec	0.061	0.039	1.538	0.125
2021 $\times$ Toronto	-0.037	0.077	-0.482	0.630
2021 $\times$ Natural	0.010	0.031	0.329	0.742
Montreal $\times$ Natural	-0.085	0.076	-1.115	0.291
Ottawa $\times$ Natural	0.076	0.078	0.967	0.359
Quebec $\times$ Natural	-0.178	0.078	-2.281	0.050
Toronto $\times$ Natural	0.187	0.099	1.889	0.072
2021 $\times$ Montreal $\times$ Natural	-0.055	0.050	-1.099	0.274
2021 $\times$ Ottawa $\times$ Natural	0.021	0.040	0.514	0.608
2021 $\times$ Quebec $\times$ Natural	0.005	0.039	0.129	0.898
2021 $\times$ Toronto $\times$ Natural	0.029	0.077	0.375	0.708

C) Estimated marginal means contrasts (Urban - Natural)				
Urban - Natural contrast	Inter-Tegular Distance (ITD, mm)			P
	Estimate	t	df	
Quebec	0.45	2.599	7.21	0.138
Montreal	0.264	1.584	9	0.443
Ottawa	-0.056	-0.324	7.45	0.754
Toronto	-0.279	-1.141	28.29	0.526

We tested whether bumblebee inter-tegular distance (ITD) was explained by Year (2021 or 2022), Region (Quebec, Montreal, Ottawa, or Toronto), Habitat (Urban or Natural), and all interactions between these factors, using linear mixed models. **A)** Result of the Wald chi-square test **B)** Coefficients of the model. P-values are computed using Kenward-Roger degrees of freedom **C)** We estimated the effect of habitat on inter-tegular distance for each region using post-hoc tests comparing estimated marginal means contrasts. We corrected for multiple comparisons using a holm correction. Note that the P-values are computed with type II sum of squares in A and with type III sum of squares in B.

Table S6. Analyses of the sepal size and corolla surface variation in the field.

A) Wald χ^2 test, male only								
Effect	Sepal Size			P	Corolla Surface			P
	χ^2	df			χ^2	df		
Year	8.386	1		0.004	33.598	1		<0.001
Region	3.835	3		0.280	37.238	3		<0.001
Habitat	0.344	1		0.557	2.191	1		0.138
Year $\times$ Region	28.021	3		<0.001	25.117	3		<0.001
Year $\times$ Habitat	8.146	1		0.004	54.978	1		<0.001
Region $\times$ Habitat	0.303	3		0.959	16.121	3		0.001
Year $\times$ Region $\times$ Habitat	6.980	3		0.072	10.690	3		0.013

B) Coefficients of the model, male only								
	Sepal Size				Corolla Surface			
	Estimate	Std Error	t	P	Estimate	Std Error	t	P
Global mean	1.191	0.023	52.752		1.828	0.029	62.996	
2021	0.020	0.005	3.680	<0.001	-0.068	0.012	-5.479	<0.001
Montreal	0.016	0.037	0.421	0.684	-0.100	0.048	-2.073	0.069
Ottawa	-0.078	0.040	-1.968	0.082	-0.236	0.051	-4.663	0.001
Quebec	0.024	0.040	0.605	0.561	0.220	0.050	4.369	0.002
Toronto	0.038	0.040	0.957	0.363	0.115	0.052	2.213	0.054
Natural	0.014	0.023	0.622	0.550	-0.039	0.029	-1.357	0.209
2021 $\times$ Montreal	0.027	0.010	2.751	0.006	-0.034	0.022	-1.529	0.127
2021 $\times$ Ottawa	-0.004	0.009	-0.501	0.616	-0.028	0.020	-1.409	0.159
2021 $\times$ Quebec	-0.041	0.009	-4.697	<0.001	0.103	0.020	5.177	<0.001
2021 $\times$ Toronto	0.018	0.010	1.736	0.083	-0.040	0.024	-1.703	0.089
2021 $\times$ Natural	0.015	0.005	2.662	0.008	0.085	0.012	6.868	<0.001
Montreal $\times$ Natural	0.018	0.037	0.477	0.645	0.083	0.048	1.720	0.121
Ottawa $\times$ Natural	-0.005	0.040	-0.135	0.896	0.004	0.051	0.082	0.936
Quebec $\times$ Natural	-0.016	0.040	-0.399	0.700	-0.190	0.050	-3.763	0.005
Toronto $\times$ Natural	0.003	0.040	0.082	0.937	0.103	0.052	1.972	0.079
2021 $\times$ Montreal $\times$ Natural	-0.017	0.010	-1.707	0.088	-0.045	0.022	-2.044	0.041
2021 $\times$ Ottawa $\times$ Natural	-0.010	0.009	-1.090	0.276	0.061	0.020	3.035	0.002
2021 $\times$ Quebec $\times$ Natural	0.017	0.009	2.017	0.044	-0.003	0.020	-0.142	0.887
2021 $\times$ Toronto $\times$ Natural	0.009	0.010	0.858	0.391	-0.013	0.024	-0.557	0.577

C) Estimated marginal means contrasts (Urban - Natural), male only								
Urban - Natural contrast	Sepal Size				Corolla surface			
	Estimate	t	df	P	Estimate	t	df	P
Quebec 2021	-0.060	-0.646	9.36	1.000	0.293	2.363	10.3	0.234
Quebec 2022	0.067	0.719	9.43	1.000	0.623	4.992	10.6	0.004
Montreal 2021	-0.059	-0.678	10.08	1.000	-0.167	-1.386	12.2	0.572
Montreal 2022	-0.068	-0.781	10.15	1.000	-0.007	-0.056	12.4	1.000
Ottawa 2021	-0.027	-0.291	9.45	1.000	-0.223	-1.776	10.8	0.416
Ottawa 2022	-0.008	-0.080	9.48	1.000	0.364	2.905	10.7	0.103
Toronto 2021	-0.081	-0.826	11.39	1.000	-0.271	-1.905	16.2	0.374
Toronto 2022	0.012	0.130	9.66	1.000	0.018	0.140	11.4	1.000

Table S6 (continued).

D) Wald χ^2 test, male and female						
Effect	Sepal Size			Corolla Surface		
	χ^2	df	P	χ^2	df	P
Year	17.529	1	<0.001	45.647	1	<0.001
Region	3.720	3	0.293	33.065	3	<0.001
Habitat	0.240	1	0.624	2.279	1	0.131
Year $\times$ Region	24.444	3	<0.001	25.008	3	<0.001
Year $\times$ Habitat	6.152	1	0.013	63.618	1	<0.001
Region $\times$ Habitat	0.303	3	0.959	14.239	3	0.002
Year $\times$ Region $\times$ Habitat	6.806	3	0.078	6.639	3	0.084

E) Coefficients of the model, male and female								
	Sepal Size				Corolla Surface			
	Estimate	Std Error	t	P	Estimate	Std Error	t	P
Global mean	1.189	0.022	54.991		1.856	0.030	62.500	
2021	0.024	0.005	4.687	<0.001	-0.072	0.012	-6.242	<0.001
Montreal	0.014	0.036	0.387	0.708	-0.114	0.049	-2.307	0.047
Ottawa	-0.074	0.038	-1.949	0.084	-0.218	0.052	-4.201	0.003
Quebec	0.024	0.038	0.634	0.543	0.207	0.052	4.002	0.004
Toronto	0.036	0.038	0.942	0.371	0.124	0.053	2.353	0.043
Natural	0.011	0.022	0.519	0.617	-0.040	0.030	-1.351	0.211
2021 × Montreal	0.019	0.009	2.005	0.045	-0.043	0.021	-2.069	0.039
2021 × Ottawa	-0.003	0.008	-0.361	0.719	-0.014	0.019	-0.766	0.444
2021 × Quebec	-0.037	0.008	-4.468	<0.001	0.096	0.019	5.097	<0.001
2021 × Toronto	0.021	0.009	2.264	0.024	-0.038	0.021	-1.791	0.074
2021 × Natural	0.011	0.005	2.254	0.024	0.087	0.012	7.544	<0.001
Montreal × Natural	0.016	0.036	0.453	0.661	0.085	0.049	1.730	0.119
Ottawa × Natural	-0.012	0.038	-0.316	0.760	-0.020	0.052	-0.387	0.709
Quebec × Natural	-0.010	0.038	-0.271	0.793	-0.175	0.052	-3.374	0.009
Toronto × Natural	0.006	0.038	0.157	0.879	0.109	0.053	2.075	0.068
2021 × Montreal × Natural	-0.017	0.009	-1.810	0.071	-0.041	0.021	-1.939	0.053
2021 × Ottawa × Natural	-0.006	0.008	-0.770	0.441	0.042	0.019	2.210	0.027
2021 × Quebec × Natural	0.018	0.008	2.157	0.031	0.000	0.019	0.005	0.996
2021 × Toronto × Natural	0.005	0.009	0.570	0.569	-0.001	0.021	-0.046	0.963

F) Estimated marginal means contrasts (Urban - Natural), male and female								
Urban - Natural contrast	Sepal Size				Corolla surface			
	Estimate	t	df	P	Estimate	t	df	P
Quebec 2021	-0.060	-0.672	9.48	1.000	0.255	2.005	10.4	0.358
Quebec 2022	0.057	0.630	9.44	1.000	0.604	4.756	10.4	0.006
Montreal 2021	-0.044	-0.527	10.14	1.000	-0.183	-1.501	12.1	0.636
Montreal 2022	-0.066	-0.784	10.14	1.000	0.003	0.021	11.9	1.000
Ottawa 2021	-0.009	-0.097	9.43	1.000	-0.137	-1.078	10.4	0.916
Ottawa 2022	0.012	0.131	9.44	1.000	0.378	2.975	10.3	0.094
Toronto 2021	-0.068	-0.728	10.92	1.000	-0.311	-2.234	14	0.254
Toronto 2022	-0.001	-0.009	9.43	1.000	0.034	0.266	10.3	1.000

We tested whether sepal size or corolla surface was explained by Year (2021 or 2022), Region (Quebec, Montreal, Ottawa, or Toronto), Habitat (Urban or Natural), and all interactions between these factors using linear mixed models. Because the Sex factor was associated with sepal size and corolla surface in the field, we decided to run the same models with males only (A-C) and male and female (D-F) to assess the effect of females on our results. For each test (either male only or male and female combined), we present the coefficients of the models (P-values are computed using Kenward-Roger degrees of freedom) and we estimated the effect of habitat on the sepal size and corolla surface for each region using post-hoc tests comparing estimated marginal means contrasts. We corrected for multiple comparisons using holm correction. Note that the P-values are computed with type II sum of squares in A and D and with type III sum of squares in B and E.

Table S7. Analyses of the sepal size and corolla surface variation in the common garden experiment.

A) Wald χ^2 test

Effect	Sepal size			Corolla surface		
	χ^2	df	P	χ^2	df	P
Region	2.270	3	0.518	5.114	3	0.164
Habitat	8.564	1	0.003	2.684	1	0.101
Region $\times$ Habitat	8.329	3	0.040	8.136	3	0.043

B) Coefficients of the model

	Sepal Size				Corolla Surface			
	Estimate	Std Error	t	P	Estimate	Std Error	t	P
Global mean	1.167	0.016	72.623		1.530	0.028	55.374	
Montreal	0.009	0.026	0.336	0.749	0.021	0.043	0.485	0.647
Ottawa	-0.033	0.027	-1.219	0.263	-0.077	0.047	-1.631	0.148
Quebec	0.022	0.027	0.807	0.446	0.067	0.047	1.430	0.194
Toronto	0.003	0.031	0.084	0.936	-0.012	0.053	-0.215	0.836
Natural	-0.044	0.016	-2.732	0.031	-0.035	0.028	-1.269	0.248
Montreal $\times$ Natural	-0.004	0.026	-0.140	0.893	-0.046	0.043	-1.059	0.336
Ottawa $\times$ Natural	0.052	0.027	1.900	0.099	0.076	0.047	1.617	0.151
Quebec $\times$ Natural	-0.070	0.027	-2.586	0.036	-0.105	0.047	-2.245	0.058
Toronto $\times$ Natural	0.022	0.031	0.708	0.503	0.075	0.053	1.410	0.203

C) Estimated marginal means contrasts (Urban - Natural)

Urban - Natural contrast	Sepal Size				Corolla surface			
	Estimate	t	df	P	Estimate	t	df	P
Quebec	0.227	3.700	7.29	0.028	0.281	2.617	7.78	0.126
Montreal	0.095	1.660	5.42	0.459	0.162	1.709	4.61	0.459
Ottawa	-0.015	-0.249	7.13	1	-0.082	-0.762	7.02	0.942
Toronto	0.043	0.585	6.91	1	-0.081	-0.623	6.84	0.942

We tested whether the sepal size or corolla surface was explained by Region (Quebec, Montreal, Ottawa, or Toronto), Habitat (Urban or Natural), and the interaction between these factors, using linear mixed models. **A)** Result of the Wald chi-square test **B)** Coefficients of the model. P-values are computed using Kenward-Roger degrees of freedom **C)** We estimated the effect of habitat on the sepal size and corolla surface for each region using post-hoc tests comparing estimated marginal means contrasts. We corrected for multiple comparisons using holm correction. Note that the P-values are computed with type II sum of squares in A and with type III sum of squares in B.

Table S8. Analysis of the mean responses of sepal size and corolla surface, grouped by habitats, to environment change.

A) Wald χ^2 test												
Effect	Sepal Size						Corolla size					
	20 21			20 22			20 21			20 22		
	χ^2	df	P	χ^2	df	P	χ^2	df	P	χ^2	df	P
Environment	0.916	1	0.338	0.048	1	0.837	8.860	1	0.003	36.761	1	<0.001
Habitat	2.003	1	0.156	4.052	1	0.044	1.445	1	0.229	2.493	1	0.114
Environment $\times$ Habitat	4.796	1	0.028	3.707	1	0.054	0.519	1	0.471	1.486	1	0.223

B) Coefficients of the model																
	Sepal Size						Corolla Surface									
	20 21				20 22		20 21				20 22					
	Estimate	Std Error	t	P	Estimate	Std Error	t	P	Estimate	Std Error	t	P				
Global mean	1.181	0.013	93.134		1.174	0.019	60.952		1.641	0.049	33.221		1.737	0.041	42.811	
Common Garden	-0.013	0.012	-1.106	0.289	0.001	0.013	0.095	0.926	-0.110	0.037	-3.015	0.010	-0.198	0.033	-5.956	<0.001
Natural	-0.022	0.013	-1.698	0.113	-0.018	0.019	-0.918	0.373	-0.014	0.049	-0.291	0.776	-0.080	0.041	-1.983	0.067
Common Garden $\times$ Natural	-0.025	0.012	-2.186	0.048	-0.025	0.013	-1.903	0.078	-0.026	0.037	-0.720	0.484	0.040	0.033	1.207	0.247

We tested whether the sepal size or corolla surface was explained by Environment (Field or Common Garden), Habitat (Urban or Natural), and the interaction between these factors, using linear mixed models. The comparison between the common garden and the 2021 field season represents the effect of genetic and plastic changes on the two components of flower size (in the common garden, we measure the offspring of seeds collected in the field in 2021). In 2022, we compared the same generation to one another (assessing the plasticity more than the genetic changes). Therefore, we ran separate models for 2021 and 2022. **A)** Result of the Wald chi-square test **B)** Coefficients of the model. P-values are computed using Kenward-Roger degrees of freedom. Note that the P-values are computed with type II sum of squares in A and with type III sum of squares in B.

Table S9. Analysis of the relationship between the genetic component of sepal size variation and bumblebee inter-tegular distances

Effect	Sepal size			
	Sum of Squares	df	F	P
Habitat	0.006	1	1.742	0.213
ITD	0.022	1	6.531	0.026
Habitat × ITD	0.000005	1	0.001	0.971
Residuals	0.038	11		

Table S10. Analysis of the relationship between the genetic component of corolla surface variation and the visitation rate of pollinators

Effect	Corolla surface			
	Sum of Squares	df	F	P
Habitat	0.002	1	0.311	0.588
Pollinator Visitation Rate	0.086	1	13.477	0.004
Habitat × Pollinator Visitation Rate	0.096	1	14.999	0.002
Residuals	0.070	11		

Table S11. Analysis of the flower shape variation in the common garden experiment and in the field.

A) Procrustes ANOVA on flower shape, excluding the "environment" factor.							
	Factors	Df	MS	Rsq	F	P	
Sepal	Common garden	Size	1	0.48813	0.03662	5.7409	0.002
		Sex	1	0.09369	0.00703	1.1019	0.350
		Region	3	0.10132	0.02280	1.1916	0.350
		Habitat	1	0.07298	0.00547	0.8583	0.502
		Region×Habitat	3	0.08595	0.01934	1.0108	0.454
		Habitat×Population	7	0.08503	0.04465	1.0000	1.000
	Field	Size	1	3.01023	0.04240	6.6537	0.007
		Year	1	0.17214	0.00242	0.3805	0.854
		Sex	1	0.27642	0.00389	0.6110	0.695
		Region	3	1.74981	0.07394	3.8677	0.005
		Habitat	1	1.69706	0.02390	3.7511	0.019
		Region×Habitat	3	0.56702	0.02396	1.2533	0.337
	Habitat×Population	9	0.45242	0.05735	1.0000	1.000	
	Corolla	Common garden	Size	1	1.37494	0.10318	13.5472
Sex			1	0.49330	0.03702	4.8604	0.007
Region			3	0.03944	0.00888	0.3886	0.980
Habitat			1	0.04430	0.00332	0.4365	0.837
Region×Habitat			3	0.04251	0.00957	0.4189	0.972
Habitat×Population			7	0.10149	0.05331	1.0000	1.000
Field		Size	1	2.07279	0.04042	8.8874	0.002
		Year	1	0.42192	0.00823	1.8090	0.154
		Sex	1	0.52744	0.01029	2.2615	0.104
		Region	3	0.55914	0.03271	2.3974	0.036
		Habitat	1	0.20683	0.00403	0.8868	0.472
		Region×Habitat	3	0.38019	0.02224	1.6301	0.156
Habitat×Population		10	0.23323	0.04548	1.0000	1.000	

B) Procrustes ANOVA on corolla surface only, including the "environment" factor						
Factors	Df	MS	Rsq	F	P	
Size	1	4.0716	0.0605	17.4631	0.001	
Year	1	0.8022	0.01192	3.4406	0.043	
Sex	1	0.6576	0.00977	2.8204	0.059	
Region	3	0.5287	0.02357	2.2674	0.061	
Habitat	1	0.3917	0.00582	1.6800	0.162	
Environment	1	1.4451	0.02147	6.1981	0.002	
Region×Habitat	3	0.2377	0.0106	1.0194	0.424	
Habitat×Population	9	0.2332	0.03118	1	1	

A) We used Procrustes ANOVA (PERMANOVA) to assess the effect of habitat on flower shape. We constructed a model with Size, Year, Sex, Region, Habitat, and the interactions between these factors as explanatory variables of sepal or corolla shape. We omitted the Year term for the common garden as we ran the experiment once. We compute results using type I Procrustes ANOVA, meaning that the effect of Habitat is assessed after all other factors. We also added population nested in habitat in the model; this way, we were able to define the error of the ANOVA as the mean square of population nested in habitats instead of residuals, therefore considering populations as the statistical individual. This conservative approach offers us more confidence in detecting differences between habitats. **B)** We assessed the effect of the environment factor (Field vs Common Garden) on corolla shape using the same rationale as in **A)**